LAKE WATKINS

AI Game Designer * AGI Research Engineer * Technical & Creative Director

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SUMMARY

Co-Founder and Chief Architect of Magi AGI, building neurosymbolic Artificial General Intelligence (AGI) agents and an AI Game Master platform on the Hyperon cognitive architecture. 20+ years in technology (professional work since 2004; in the games industry since 2007) spanning AI research (cognitive architectures, Large Language Model (LLM) agents, conversational AI for Hanson Robotics' Sophia), engineering leadership (Technical Director, CTO, 21+ shipped game releases, 12+ patents), and creative game design (30+ years GMing, 10+ years as a professional GM, with longitudinal player-experience data). Equally credible as a senior game designer / creative director and as a founding / staff engineer at AI startups. Specializes in the intersection of Large Language Models, symbolic AI (MeTTa / Atomspace / Hyperon), and interactive narrative systems.

CORE SKILLS & TECHNOLOGIES

Artificial Intelligence & Research: Neurosymbolic AI * Artificial General Intelligence (AGI) * Large Language Model (LLM) agents * Cognitive Architectures * Hyperon * MeTTa * Atomspace * OpenPsi * Probabilistic Logic Networks (PLN) * Goal Management * Causal Reasoning * Temporal Credit Assignment * Knowledge Graphs * Hypergraph Reasoning * Behavior Trees * Hierarchical Task Networks (HTN) * Reinforcement Learning * AI Alignment * Interpretability * Conversational AI * Dialogue Systems * Action Planning * Multi-Agent Systems

Game Design & Direction: Game Design * Systems Design * Narrative Design * Level Design * Technical Design * Creative Direction * Gameplay Programming * Game Prototyping * Game Balancing * Live-Service Design * AI-Driven NPCs * Procedural Content * Tabletop / Role-Playing Game (RPG) Design * Game-Based Learning * Educational Game Design * Player Retention Analysis

Engineering: C++ * C# * C * Python * Java * Objective-C * JavaScript * Unity * Unity3D * Unreal Engine (familiar) * Cocos2D * Shaders (HLSL / GLSL) * Fluid Simulation * Real-Time Rendering * Mobile Development (iOS, Android) * Cross-Platform Engines * Git * Version Control * Continuous Integration / Continuous Deployment (CI/CD) * Agile * Scrum

Leadership & Operations: Technical Leadership * Engineering Management * R&D Direction * Cross-Functional Team Building * Hiring & Mentorship * Stakeholder & Investor Relations * Pitch Authorship * Whitepaper Authorship * Grant Writing * Open-Source Strategy * Patent Authorship

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WORK EXPERIENCE

Co-Founder & Chief Architect - Magi AGI
2025 - Present * San Francisco, CA

- Co-founded Magi AGI to build interpretable, goal-directed AI systems for interactive worlds - proving the architecture through game products with an AI Game Master as the lead product.
- Architected MAGUS (Modular Adaptive Goal and Utility System), a hierarchical goal-management framework with an overgoal mechanism that evaluates goal fitness by measurability and inter-goal correlation, preventing instrumental convergence through explicit separation of metric measurement from goal-driven behavior.

- Architected HERMES (Hypergraph Experiential Reasoning & Motivational Engagement System), a causal-knowledge-graph engine performing temporal credit assignment over experiential traces and embedding causal attributions in MeTTa / Atomspace for goal-directed planning.
- Designed the closed-loop integration between HERMES, MAGUS, and OpenPsi action selection - experience -> causal structure -> goal prioritization -> action -> new experience.
- Direct architecture for the AI Game Master product: a neurosymbolic agent targeting professional-quality tabletop campaign play over 700+ hours per playthrough, benchmarked against a 10-year, ~170-session reference dataset of longitudinal player-retention data at $80/month.
- Designed the dual-entity corporate structure (Magi AI, Inc. C-Corp + Magi Games Cooperative LLC) balancing venture-scale AI research with cooperative talent retention for creative work.
- Lead cross-disciplinary team across cognitive architecture, game design, narrative, and product; manage grant strategy (NSF SBIR, AI Safety / Knight / Mozilla funder pipeline).

Steward of Design - Glass Umbrella
01/2025 - Present

- Direct design and development on an unannounced AI-driven game and educational project.
- Run multi-year tabletop RPG campaigns with multiple concurrent weekly groups; this practice generates the longitudinal player-experience data that anchors the Magi AI Game Master quality bar.

Neoterics Lead - SingularityNet
06/2024 - 04/2025

- Led the Neoterics initiative building first-generation AGI agents on the PRIMUS / Hyperon neurosymbolic cognitive architecture.
- Designed conversational AIs combining LLM-based improvisation with authored narrative scripts and meaningful player choice; shipped integrations into the SophiaVerse game.
- Coordinated cross-functional engineering between SingularityNet’s AI platform team and external partner game teams.

Partner Engineer - Take-Two Interactive Software, Inc.
11/2022 - 05/2024

- Directly supported a dozen-plus internal game teams integrating Take-Two’s social and platform services.
- Defined partner-lifecycle and engagement metrics adopted as the internal partner-tracking standard.
- Built reference demos of the social-services SDK on a new platform for partner onboarding and training.

Co-Founder & CTO - Hark
09/2021 - 08/2022 * Game Development Studio

- Designed and developed game prototypes including a fluid-and-life simulation shader system.
- Managed cross-disciplinary team across engineering, art, music, design, web development, and marketing.
- Closed investors, partners, and team hires; authored the pitch deck, game design document (GDD), trailer script, whitepaper, and other foundational documentation.

Founder & Game Master - Ledge Foundation
08/2015 - 12/2021 * Open Gaming Foundation / Game-Based Learning Social Enterprise

- Founded and ran a multi-year organized-play foundation; designed, developed, and shipped multiple complete tabletop game experiences including narratives, characters, environments, combat encounters, and meta-systems.
- Designed full English- and Mathematics-education curricula delivered through gameplay; ran live sessions in partnership with schools and educators.

- Built and sustained a dedicated player community with 90%+ month-to-month retention sustained across a decade of professional GMing; the underlying retention data (~170-session campaigns over 10+ years) is now the empirical benchmark for the Magi AI Game Master product.

AI Software Designer - SingularityNET / Hanson Robotics
04/2018 - 05/2019

- Designed the goal- and motivational-system architecture for the Sophia humanoid robot - personality, dialogue, and action planning.
- Led the design of the GHOST scripting language for chatbot authoring and improved dialogue control.
- Coordinated AI, hardware, and creative teams to align organizational goals and product strategy.

Vision Designer & Team Lead - Hong Kong Polytechnic University
09/2016 - 04/2018

- Designed, developed, and released Ilolo, a mobile game for children with special educational needs.
- Owned the core vision design and implemented the first prototypes.
- Oversaw focus testing in local schools with direct feedback from players, parents, and educators.

Lead Game Designer / Developer - OpenCog Foundation
02/2012 - 06/2015

- Designed and developed a Minecraft-style game world as a training-and-evaluation environment for embodied AGI agents.
- Built new AI systems including a behavior-tree framework integrated with a hierarchical planning architecture.
- Extended Unity3D as an AGI host environment in C#, C++, Java, and Python.

Technical Director / R&D Lead / Mobile Game Developer - XMG Studio
11/2009 - 11/2011

- Shipped 21+ game releases across 9 franchises on iOS, Android, and Windows Phone 7.
- Programmed extensively in Objective-C, C++, and C#; built on Unity3D and Cocos2D.
- Managed multiple R&D teams; hired and onboarded 20+ engineers.
- Co-invented and patented 12+ innovations in mobile game design.

SELECTED RESEARCH

- MAGUS and HERMES: A Two-Way Bridge for Experience-Grounded Goal Management in Neurosymbolic Agents - co-authored with Anna Mikeda and Ursula Addison (Magi AGI). Architecture and formalism for a closed-loop neurosymbolic motivational system on the Hyperon stack; integration with OpenPsi action selection; evaluation framework against LLM-only and pure-symbolic baselines. (Manuscript in submission; full publication detail will be added on acceptance.)

SELECTED RECOMMENDATIONS

> "Lake proved again and again his dedication to getting the job done right and thinking things all the way through. He is an amazing architect and has a passion for in-depth game systems design that is a rare craft these days." - Adam Telfer, Senior Director of Product, Bungie

> "He is the best programmer I have ever had the chance to work with. Not only was he technically knowledgeable, he was also an effective communicator and a patient teacher ... a rare mix of skill, diligence and great personality." - Sagar Patel, Freelance Creative Coder / Experience Designer

> "A brilliant programmer always ready to put his energy into getting the job done. Precise and deadline oriented. A wise team player who can be trusted." - Adam Cohen, UI/UX Engineer, Autodesk

> "His knowledge in design and project management makes the projects he runs go smoothly ... Lake also goes beyond the call of his position bringing a high moral and sense of comradery to the work environment." - Rocco Commisso, Game Designer, Tech Artist & Instructor

Additional references available on request.